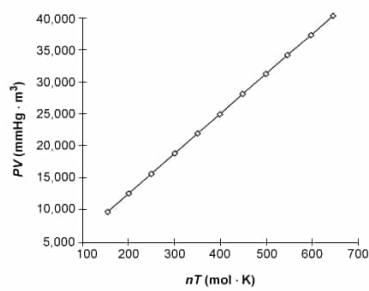


The product of pressure P and volume V of an ideal gas is directly proportional to the product of the moles n and absolute temperature T of the gas, as shown below.

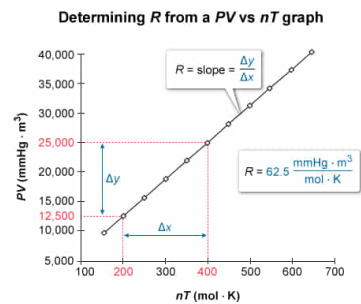


What is the value of the gas constant R derived directly from the graph of the data?

- ☐ A. 0.0160 mol·K/mmHg·m³ [3%]
- ☐ B. 0.0821 L·atm/mol·K [17%]
- ☐ C. 8.31 J/mol·K [4%]
- ☒ D. 62.5 mmHg·m³/mol·K [74%]

Correct
73% answered correctly

Explanation:



An **ideal gas** is a hypothetical gas with idealized **characteristics** used to make a simplified model to predict the behavior of real gases. According to this model, the **average kinetic energy KE** of a single gas molecule is directly proportional to temperature via the **Boltzmann constant** ($k = 1.381 \times 10^{-23}$ J/K). When the Boltzmann constant is expressed *per mole* of a gas, it is known as the **gas constant** ($R = 8.31$ J/mol·K):

$$\left(1.381 \times 10^{-23} \frac{\text{J}}{\text{K}} \right)$$

Because pressure correlates to **energy per unit volume**, the joules unit of R can be converted to units of pressure and volume. For example, expressing R in units of liters and atmospheres gives:

$$R = 8.31 \frac{\text{J}}{\text{mol} \cdot \text{K}}$$

As a result, the exact numerical value of R depends on the units used to measure it. Accordingly, when calculations with R are performed, the value of R used must match the units in the calculation. For example, the behavior of a gas can be described by the **ideal gas law**:

$$P V = n R T$$

where P is the pressure, V is the volume occupied, n is the number of moles, and T is the absolute temperature (in Kelvin). This relationship shows that PV is directly proportional to nT , and R is the proportionality constant:

$$R = \frac{P V}{n T}$$

Therefore, a graph of PV (y -axis) vs. nT (x -axis) results in a straight line with a slope of R .

Evaluating the slope of the line from the graph given in this question directly yields the value of R without requiring any unit conversions:

$$R = \text{slope} = \frac{\Delta y}{\Delta x}$$

$$R = \frac{\Delta y}{\Delta x}$$

(Choice A) $0.0160 \text{ mol}\cdot\text{K}/\text{mmHg}\cdot\text{m}^3$ results from calculating the reciprocal slope ($\Delta x/\Delta y$), instead of the slope ($\Delta y/\Delta x$), of the line.

(Choices B and C) Although $0.0821 \text{ L}\cdot\text{atm}/\text{mol}\cdot\text{K}$ and $8.31 \text{ J}/\text{mol}\cdot\text{K}$ are valid values of R (expressed in alternate units), they are not the value of R derived directly from the graph expressed in the units of the given data.

Educational objective:

The average kinetic energy of a single ideal gas molecule is directly proportional to temperature via the Boltzmann constant, which can be expressed per mole of gas as the ideal gas constant R . According to the ideal gas law, PV is directly proportional to nT , and a graph of PV vs. nT results in a straight line with a slope of R .

Time spent: 0 sec

Subject:
General Chemistry

QID: 402621

System:
Thermodynamics, Kinetics & Gas Laws